

- 4.6 Demonstrate an understanding of the idea of linear momentum conservation
- 4.7 Demonstrate an understanding of the idea of rate of change of momentum to explain protective features including bubble wraps, seat belts, crumple zones and air bags
- 4.8 *Investigate how crumple zones can be used to reduce the forces in collisions*

- 4.9 **Use the equation:**

force (newton, N) = change in momentum (kilogram metre per second, kg m/s) / time (second, s)

$$F = \frac{(mv - mu)}{t}$$

to calculate the change in momentum of a system, as in 4.6

- 4.10 Use the equation:

work done (joule, J) = force (newton, N) × distance moved in the direction of the force (metre, m)

$$E = F \times d$$

- 4.11 Demonstrate an understanding that energy transferred (joule, J) is equal to work done (joule, J)

- 4.12 Recall that power is the rate of doing work and is measured in watts, W

- 4.13 Use the equation:

power (watt, W) = work done (joule, J) / time taken (second, s)

$$P = \frac{E}{t}$$

- 4.14 Recall that one watt is equal to one joule per second, J/s

- 4.15 Use the equation:

gravitational potential energy (joule, J) = mass (kilogram, kg) × gravitational field strength (newton per kilogram, N/kg) × vertical height (metre, m)

$$\text{GPE} = m \times g \times h$$

- 4.16 Use the equation:

kinetic energy (joule, J) = $\frac{1}{2} \times$ mass (kilogram, kg) × velocity² ((metre/second)² (m/s)²)

$$\text{KE} = \frac{1}{2} \times m \times v^2$$

- 4.17 Demonstrate an understanding of the idea of conservation of energy in various energy transfers

- 4.18 **Carry out calculations on work done to show the dependence of braking distance for a vehicle on initial velocity squared (work done to bring a vehicle to rest equals its initial kinetic energy)**